STARTUP PROFIT ANALYSIS

Analyzing the Impact of R&D, Administration & Marketing Spend on Profit

R&D Innovation | Administration Operations | Marketing Growth | Profit Success

State: All

Total Startups	Total Profit	Average Profit	Average R&D Spend
50	5.60M	112.01K	73.72K

KEY INSIGHTS

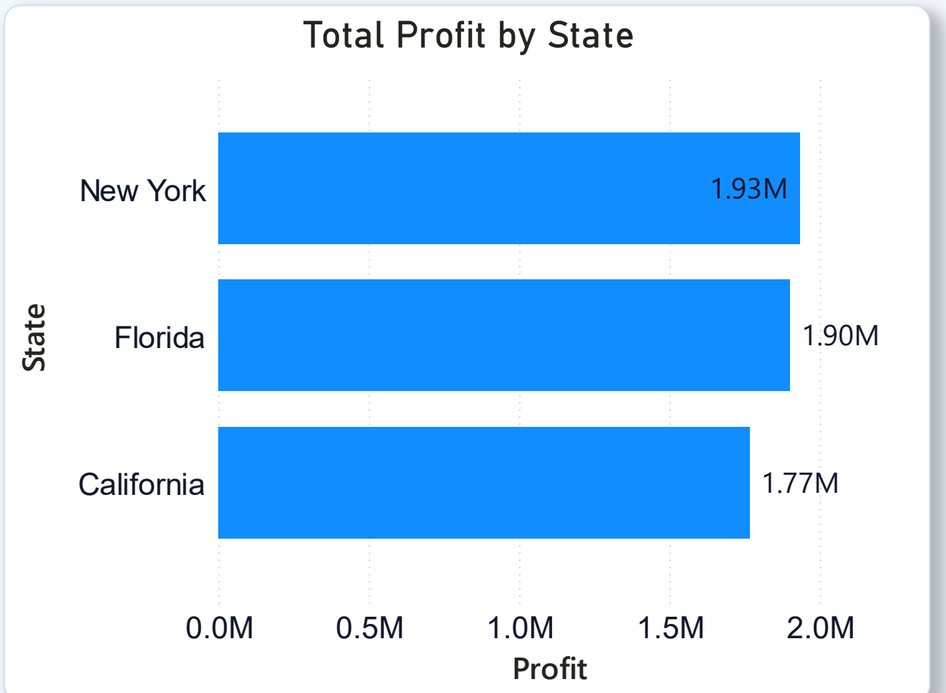
1. State-wise Profit
Compare total profit across California, Florida, and New York to identify the leading state.

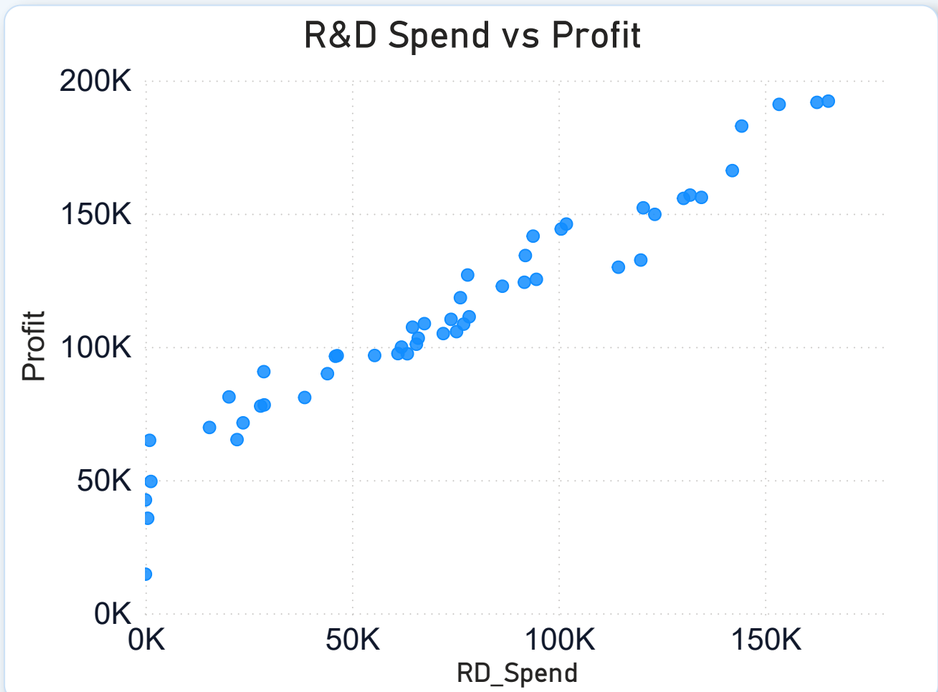
2. R&D Spend
The regression model shows a positive relationship between R&D spending and predicted profit, holding other spending variables constant.

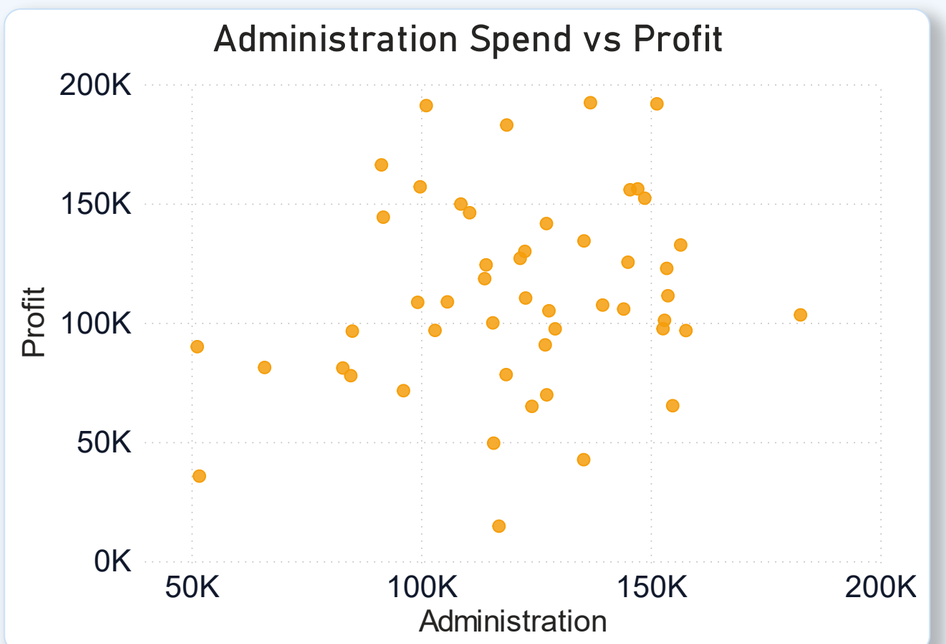
3. Administration Spend
The regression coefficient for administration spending is negative. This indicates a negative modelled relationship when the other spending variables are held constant.

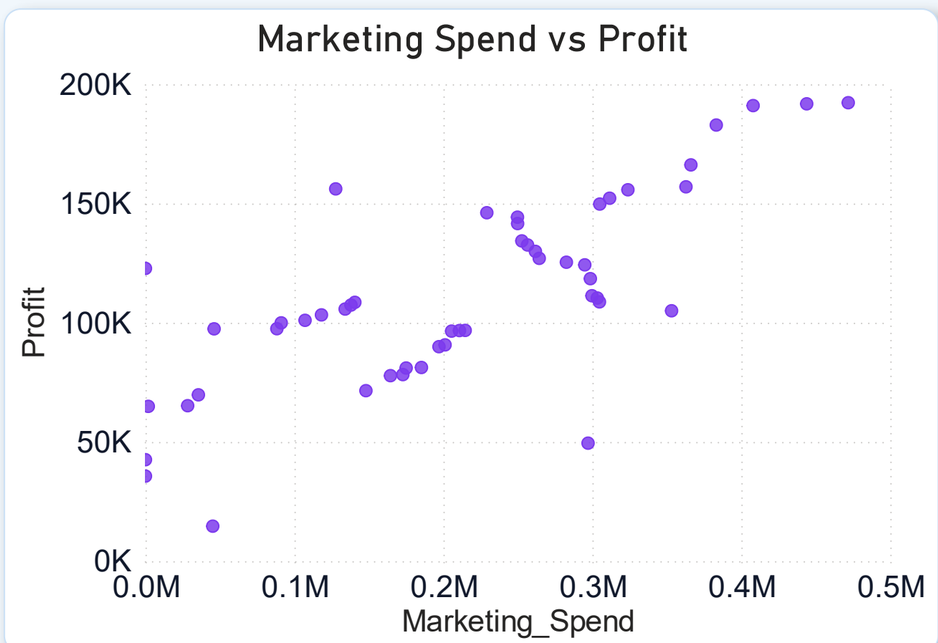
4. Marketing Spend
The regression coefficient for marketing spending is positive, but the relationship should be interpreted carefully.

5. Model Performance
The model achieved an R^2 score of approximately 0.9001 and an MAE of approximately 6,979.15 on the test dataset.









REGRESSION MODEL SUMMARY

R^2 Score: 0.9001

Mean Absolute Error: 6,979.15

Profit = 54071.88 + (0.8038 × RD Spend) − (0.0679 × Administration) + (0.0312 × Marketing Spend)